

DS3040 Introduction to Deep Learning Lab

Assignment-2, Date: Jan 13, 2025

Timing: 10:00 AM to 12:45 PM

Max Marks: 20

Instructions

1. Submit one .ipynb file containing all answers. The name should be [student_name]_assignment[number].ipynb
 2. Write the questions in separate text blocks before the answers.
 3. Outputs for all sub-questions should be given and the code should be executable.
 4. Write justifications for your choices where needed.
 5. Ensure that all plots include clear labels and legends for better interpretation.
 6. Do not use AI or any websites except library documentations.
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1. Perceptron for Binary Classification

(5)

Implement the Perceptron algorithm for binary classification.

- a Generate a linearly separable dataset with 5 features, 2 classes, and 1000 samples using sklearn's *make_classification*.
- b Train your Perceptron model on this dataset and experiment with different learning rates (e.g., 0.01, 0.1) and iterations (e.g., 50, 100).
- c Plot the decision boundary for the best model on training data and evaluate the model on test data.

2. Perceptron for Multi-Class Classification

(5)

Modify your Perceptron implementation to handle multi-class classification.

- a Load the MNIST dataset from *sklearn.datasets* and preprocess it after selecting a random subset of 10,000 samples.
- b Train the model and evaluate its performance on the test dataset.

3. Logistic Regression

(5)

Modify your algorithm to predict probabilities instead of discrete outputs.

- a Train the modified model on the same MNIST dataset.
- b Evaluate the model on the same test set and report the accuracy and classification metrics.

4. Comparison

(5)

Compare your Logistic Regression implementation with sklearn's.

- a Train LogisticRegression from sklearn on the same MNIST dataset.
- b Compare the performance (accuracy, precision, recall, F1-score) and training time with your implementation.